



Different Laws in Chemistry

Boyle's Law



"At constant temperature, the volume of a gas is inversely proportional to the pressure applied"

$$PV = k$$
$$P_1 V_1 = P_2 V_2$$

Named after the British Chemist, Robert Boyle

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Charles' Law



"At constant pressure, the volume of a gas is directly proportional to the absolute temperature."

$$\frac{V}{T} = k$$
$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Named after the French Physicist, Jacques Alexandre Charles

Gay-Lussac Law

"At constant volume, the pressure of a gas is directly proportional to the absolute temperature."

$$\frac{P}{T} = k$$
$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$



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Named after the French Chemist, Joseph Louis Gay-Lussac

Amontons' Law

"If the volume of a gas is held constant, increasing the temperature of the gas increases its pressure"



Named after the French physicist, Guillaume Amontons. This is just the same as Gay-Lussac's Law

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Ideal Gas Law

Combining the gas laws will result in a single equation, called the **ideal gas law**. This is the equation that relates all the four variables (P, V, n and T). All gases whose behavior is approximated by this equation are called ideal gases. In this equation, the proportionality constant R is called the gas constant.

$$PV = nRT$$

$$R = 0.0821 \text{ L}\cdot\text{atm/mol}\cdot\text{K}$$

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General Gas Law

Changing states (changing the values of pressure, volume and temperature) will change the ideal gas law to the combined gas law or the general gas law.

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Avogadro's Law



"The volume of a gas and the number of moles present is directly proportional at constant temperature and pressure."

$$\frac{V}{n} = k$$

Named after the Italian scientist, Lorenzo Amadeo Avogadro

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Dalton's Law of Partial Pressures

"The total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of all gases in the mixture."

$$P_{\text{total}} = P_1 + P_2 + \dots + P_n$$

Also known as Dalton's Law, named after the English Chemist, John Dalton

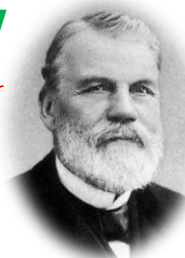


Raoult's Law

"When a nonvolatile solute is added to water or another solvent, the vapor pressure of the solvent is depressed."

$$P_{\text{soln}} = X_{\text{solvent}} P^{\circ}_{\text{solvent}}$$

Named after French chemist François-Marie Raoult



Henry's Law

"The solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid."

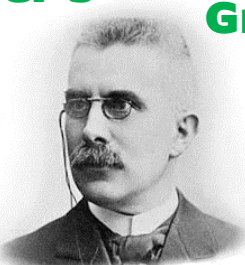
$$S = k_H P$$

Named after the English Chemist, William Henry



Le Chatelier's Principle

"If a stress is imposed on a reaction at equilibrium, the reaction will shift either left or right in the way or direction that accommodates or relieves the stress."



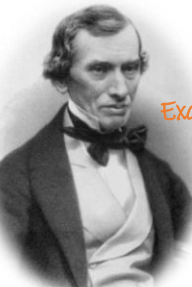
Also known as the "Equilibrium Law", named after the French Chemist, Henry Louis Le Chatelier

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Graham's Law of Effusion

"The rates of effusion of two different escaping, A and B, are inversely proportional to the square roots of their molar masses."

$$\frac{\text{Effusion rate}_A}{\text{Effusion rate}_B} = \sqrt{\frac{MM_B}{MM_A}}$$



Named after the Scottish Chemist, Thomas Graham

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Pauli Exclusion Principle

"In an atom or molecule, no two electrons can have the same four electronic quantum numbers."

Named after the Austrian Physicist, Wolfgang Pauli

Periodic Law

"If the chemical elements are listed in order of increasing atomic number, many of their properties go through cyclical changes, with elements of similar properties recurring at intervals." This principle was discovered by Russian chemist Dmitri Mendeleev

